



Syrine Elayeb

Full Stack Engineer

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Profile

Full Stack Engineer with internship experience developing AI-powered, IoT-enabled web applications and distributed systems. Skilled in React, Node.js, Spring Boot, MQTT, Docker, and RESTful APIs, with a track record of building scalable full-stack solutions from design to deployment.

Experience

Full Stack Engineer Intern

Feb 2026 – Jun 2026

QuetraTech, Mahdia

HaulTrack — Intelligent Fleet Management Platform

- Led end-to-end development as the sole engineer across 4 Agile sprints and 2 production releases, owning system design, implementation, and deployment.
- Architected a RESTful Node.js/Express + MongoDB backend spanning 13 modules and 42+ API endpoints, integrating Redis for live GPS caching and MQTT for IoT device telemetry.
- Designed and deployed a YOLOv8 + EasyOCR microservice via FastAPI for automated license plate recognition, validating 6 Tunisian plate formats from real-time RTSP camera streams with duplicate-detection suppression logic.
- Implemented JWT-based authentication and role-based access control across 2 user roles (Admin, Shipment Manager) for secure multi-user operations.
- Delivered a React + Tailwind fleet dashboard (44+ Vercel deployments) with live GPS tracking, geofencing, mission monitoring, and speed-violation alerts via Socket.IO.
- Collaborated in an Agile Scrum environment with Product Owner and Scrum Master using Git workflows and iterative delivery.

Full Stack Developer Intern

Feb 2024 – May 2024

LeaderPos

E-Learning Platform

- Built a full-stack e-learning platform supporting course management, meeting scheduling, and 3 content types (video, PDF, quizzes), with automated email notifications for enrollments.
- Engineered RESTful APIs using Spring Boot and Java to support a scalable, modular backend architecture.
- Integrated Angular frontend with Spring Boot services to deliver a responsive, seamless user experience.
- Modeled normalized MySQL database schemas to ensure data consistency, scalability, and optimized query performance.
- Contributed to testing, debugging, and continuous feature delivery across the software development lifecycle.

Projects

RoboHealth — AI Robot Health Monitoring System

Jan 2026 – Feb 2026

Stack: React | TypeScript | Node.js | MQTT | XGBoost | Docker

Real-time predictive maintenance system for industrial robots using IoT telemetry and machine learning.

- Built a microservices monitoring platform (React/TypeScript frontend, Node.js/Express backend, MongoDB, MQTT broker) orchestrated with Docker Compose across 5 containerized services, including a Python robot simulator.
- Trained an XGBoost model to detect 3 failure patterns — battery degradation, abnormal velocity, and motor overheating — from real-time telemetry spanning 16+ robot, sensor, and embedded-PC metrics.
- Implemented JWT authentication with role-based dashboards for Admin and Maintenance Engineer roles, delivering real-time anomaly alerts via Socket.IO.

Technical Skills

Programming Languages	JavaScript, TypeScript, Python, Java, SQL
Frontend	React.js, Angular, Vite, Tailwind CSS, HTML5, CSS3
Backend	Node.js, Express.js, FastAPI, Spring Boot, RESTful APIs
Databases	MongoDB, MySQL, Redis
IoT & Communication	MQTT, Socket.IO, ESP32, GPS, Telemetry
AI & Computer Vision	XGBoost, Scikit-learn, YOLOv8, OpenCV, Machine Learning, Anomaly Detection
DevOps & Tools	Docker, Git, GitHub Actions, CI/CD, Vercel, Render, Postman

Education

Master's Degree in Internet of Things	2024 – 2026
Higher Institute of Technological Studies of Mahdia (ISETMA)	
Bachelor's Degree in Information Systems Development	2021 – 2024
Higher Institute of Technological Studies of Mahdia (ISETMA)	

Certifications

Building LLM Applications with Prompt Engineering	NVIDIA, 2025
Building Conversational AI Applications	NVIDIA, 2025
Building RAG Agents with LLMs	NVIDIA, 2025
English Proficiency Certificate — B2	EF SET, 2026
Full-Stack Development: Java, Spring Boot & Angular	PROTECH-IT, 2023

Languages

Arabic (Native) | French (Fluent) | English (B2 Upper Intermediate)